Network and IoT Security | NIoSec

Summary

CONTENTS

1. Introduction ... 3
2. TLS ... 3
2.1. Specification ... 3
2.1.1. Record protocol ... 4
2.1.2. Subprotocols ... 4
2.1.3. Handshake ... 4
2.2. Modern TLS Variants ... 4
2.2.1. DTLS ... 4
2.2.2. QUIC ... 5
2.3. Attacks ... 5
2.3.1. Heartbleed ... 5
2.3.2. Syn Flooding ... 5
2.3.3. TCP Reset ... 5
2.3.4. FREAK ... 5
3. Cryptography ... 6
3.1. Protocols ... 6
3.2. Standards ... 6
3.3. Symmetric-Key Cryptography ... 6
3.3.1. Block Cipher ... 7
3.3.2. Stream Cipher ... 8
3.3.3. Overall Construction ... 8
3.4. Asymmetric Cryptography ... 8
3.4.1. Signature Schemes ... 8
3.4.2. Public Key Infrastructure (PKI) ... 9
3.4.3. Signatures vs MACs ... 9
3.4.4. Establishing a shared key ... 9
3.4.5. Asymmetric Encryption ... 10
3.4.6. Overall Construction ... 10
3.5. Maths <3 ... 10
3.6. Quantum Cryptography ... 11
3.6.1. Quantum Key Distribution ... 11
3.6.2. BB84 (By example of polarized light) ... 11
4. IPsec ... 12
4.1. Protocols ... 12
4.1.1. Authentication Header (AH) ... 12
4.1.2. Encapsulating Security Payload (ESP) ... 13
4.2. Modes of Operation ... 14
4.2.1. Transport vs. Tunnel Mode ... 14
4.2.2. Transport Mode ... 14
4.2.3. ESP Tunnel Mode ... 14
4.3. Security Associations ... 14

4.3.1. SAD and SPD . . . 15
4.3.2. Handling of outbound packets . . . 15
4.3.3. Handling of inbound packets . . . 15
4.4. Internet Key Exchange v2 (IKEv2) . . . 15
4.4.1. Initial Exchanges . . . 16
4.4.2. AUTH . . . 16
4.4.3. Child SA Exchange . . . 17
4.4.4. INFORMATIONAL . . . 17
4.4.5. Key Derivation (Function) . . . 17
4.4.6. Algorithms . . . 18
4.5. IPsec & NAT . . . 18
4.5.1. Problems . . . 18
4.5.2. NAT-T and IPsec . . . 18
5. WireGuard . . . 19
5.1. Crypto stack . . . 19
5.2. Setup . . . 19
5.3. Protocol Phases . . . 19
5.3.1. Connection establishment . . . 19
5.3.2. First Handshake Message . . . 19
5.3.3. Second Handshake Message . . . 19
5.3.4. Key Derivation . . . 20
5.4. IKEv2/IPsec vs WireGuard . . . 20

1. INTRODUCTION

OSI Stack:

Communication layers	*Security protocols*
Application layer	Web Application Security, VoIP Security, SW Security
Transport layer	TLS
Network layer	IPsec
Data Link layer	L2TP, IEEE 802.1X, IEEE 802.1AE, IEEE 802.11i (WPA2)
Physical layer	Quantum Key Exchange

The CIA triad is a foundational information-security model stating that systems should protect:

Confidentiality: Keeping information secret

Integrity: Keeping information correct and unaltered

Availability: Ensuring information and systems remain accessible

Additional desired properties include:

Authenticity: Adversary cannot claim to be someone else

Accountability/Non-repudiation: Actions can be traced to actor

2. TLS

SSL and TLS are protocols for internet handshakes and encrypted transmission. Secure Socket Layer (SSL) came first, then after v3.0 it became Transport Layer Security (TLS), currently v1.3. People still use SSL as a term, even though technically it's now TLS.

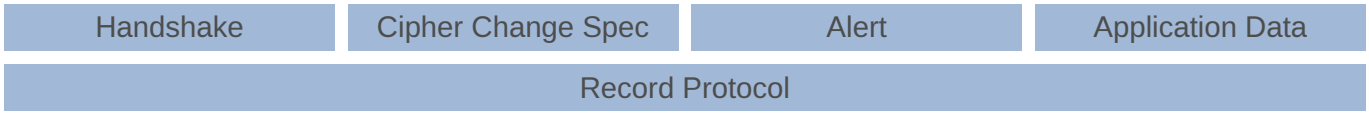
TLS 1.2: Introduced authenticated encryption, which is the most modern form of encryption. It's a more flexible protocol, lots of TLS extensions, the hashing function has been improved.

TLS 1.3: The protocol has been redesigned to make it faster. A lot of the old ciphers have been removed to make it more secure. It only supports authenticated encryption (AEAD).

TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256
TLS_DHE_RSA_WITH_AES_128_GCM_SHA256
TLS_DHE_RSA_WITH_AES_256_GCM_SHA384

Encryption_Key protocol	_Key exchange algorithm	_Signature_WITH_ algorithm	Bulk encryption algorithm	_Message Authentication Code (MAC)

2.1. SPECIFICATION



2.1.1. Record protocol

All TLS packets are sent using the record protocol. This header is attached to every single message.

Header			Data
Type (1B)	Version (2B)	Length (2B)	…

Type: which kind of TLS message is sent (20,21,22,23)

Version: which TLS version is used

Length: how long is the payload of the message

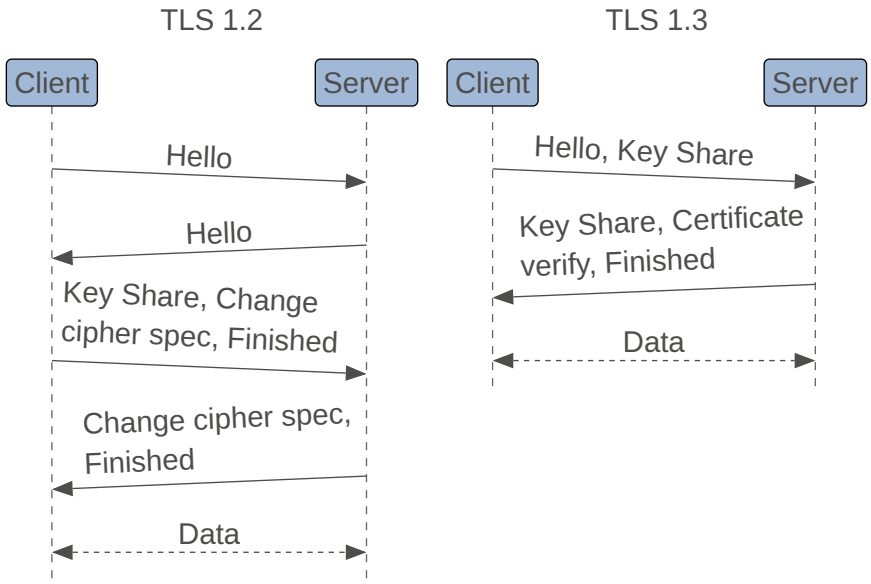
2.1.2. Subprotocols

Protocol	*Definition*
20 ChangeCipherSpec	The sender has sufficient information and is switching to encryption or to a new cipher suite (sent very rarely). By starting encryption for the first time, it is saying that the handshake has been successful, and encryption can start
21 Alert	An SSL-level alert notification (example close_notify). Not necessarily an error. Example: problem with the handshake, received data length is wrong, warning.
22 Handshake	Messages sent right at the beginning (example ServerKeyExchange). Eg. To establish a cipher, to establish our keys
23 Application Data	Opaque application data. Any applications data once we start the encryption is going to be sent under protocol 23

2.1.3. Handshake

The TLS handshake protocol is used to establish parameters for the remainder of the session. It must:

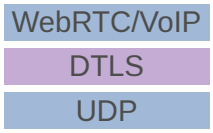
– Agree ciphers and protocols
– Establish shared secrets
– Authenticate server and client
– Be robust to tampering and attacks



2.2. MODERN TLS VARIANTS

2.2.1. DTLS

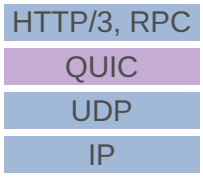
– TLS on top of UDP
– Used for WebRTC, VoIP
– Does not provide ordering of packets
– Packets may drop, retransmission is delegated to application





2.2.2. QUIC

– TLS 1.3 + UDP
– Provides reliable communication
  – Stateful
  – Multiple streams
– Used for HTTP/3, RPC



2.3. ATTACKS

2.3.1. Heartbleed

The Heartbeat extension is a keep-alive feature of TLS. The `heartbeat_request` message includes payload length, payload and padding fields. The Heartbleed bug is an implementation flaw in that extension.

1) Send message that indicates the maximum payload length (64 KB) that only includes the minimum payload (16 bytes)
2) Receive almost 64 KB of random memory that the server returns
3) Profit (look for private keys, auth cookies, etc.)

Countermeasures: Update OpenSSL

2.3.2. Syn Flooding

Idea: Fill the TCB (Transmission Control Block) queue storing the half-open (never ACKed) TCP connections so that there will be no space to store TCB for any new half-open connections. Basically the server cannot accept any new SYN packets.

1) Use random source IP addresses; otherwise the attacks may be blocked by the firewalls.
2) Spam TCP requests
3) Profit (sleep well knowing you're a menace to society)

The SYN+ACK packets sent by the server may be dropped because forged IP address may not be assigned to any machine. If it does reach an existing machine, a RST packet will be sent out, and the TCB will be dequeued.

Countermeasures: SYN Cookies
– After a server receives a SYN packet, it calculates a keyed hash (H) from the information in the packet using a secret key that is only known to the server.
– This hash (H) is sent to the client as the initial sequence number from the server. H is called SYN cookie.
– The server will not store the half-open connection in its queue.
– If the client is an attacker, H will not reach the attacker.
– If the client is not an attacker, it sends H+1 in the acknowledgement field.
– The server checks if the number in the acknowledgement field is valid or not by recalculating the cookie.

2.3.3. TCP Reset

Spoof RST Packet to break up a TCP connection between Alice and Bob.

The following fields need to be set correctly: Source IP address, Source Port, Destination IP address, Destination Port, Sequence number (within the receiver's window)

2.3.4. FREAK

– Man-in-the-middle (MitM) downgrades used TLS cipher suite to use export RSA
– MitM then factors 512 bit RSA to get session key

TODO:
https://ciphersuite.info/cs/TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256/

3. CRYPTOGRAPHY

Parties: The abstract entities in cryptographic protocols (servers, switches, humans)

Attacker model: Defines the capabilities of the attacker (computational limits, corruption)

Secure channel: Provides Confidentiality (we allow length leakage) and Authenticity (we don't care about liveness)

Authentic channel: Only provides Authenticity

3.1. PROTOCOLS

TODO:
W2 slides 18

3.2. STANDARDS

Don't roll your own crypto, except if you work at FAANG, they shall perish.

National Institute of Standards and Technology (NIST) publishes
– Special Publications (SP), guidelines and recommendations
– Federal Information Processing Standards (FIPS), standards for US government

Internet Society (ISOC)
– Organizes working groups for TCP/IP
– Publishes RFCs

3.3. SYMMETRIC-KEY CRYPTOGRAPHY

Insecure channel: Vulnerable to MITM Attacks

Shared key: Secure means the adversary has no information about the shared key. Establishing a shared key is non-trivial
  – Pre-shared key (Code book)
  – Quantum key exchange
  – Public Key Cryptography

Message Authentication Codes (MAC): Provides authentic channel over an insecure channel (a short piece of information used for authenticating and integrity-checking a message, like a signature but symmetric). Vulnerable to length-extension-attacks.

Cryptographic hash function: Should be collision-resistant and one-way. Good: SHA2, SHA3, Bad: MD5, SHA-1

Symmetric Encryption: Provides secure channel over an authentic channel

Hash-based MAC (HMAC): Fixes length-extension-attack vulnerability.

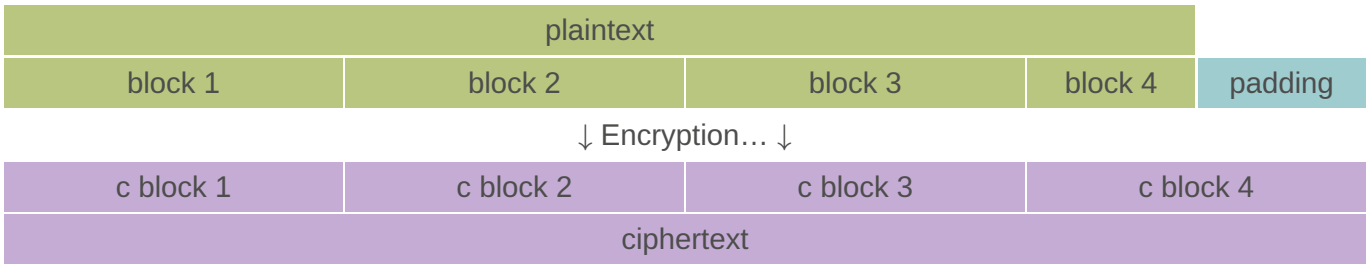
$$\mathrm{Hash}(\mathrm{key} \oplus \mathrm{opad} \mid \mathrm{Hash}\ (\mathrm{key} \oplus \mathrm{ipad} \mid m))$$

Block Size: Size of transformed/compressed data in the hash function for each iteration.

Message Digest Size: The output size of the hash function, also called hash value.

3.3.1. Block Cipher

Block ciphers take an input of a fixed size and return an output of the same size



– Block ciphers attempt to hide the transformation from message to ciphertext through confusion and diffusion
– Most block ciphers are SP-Networks

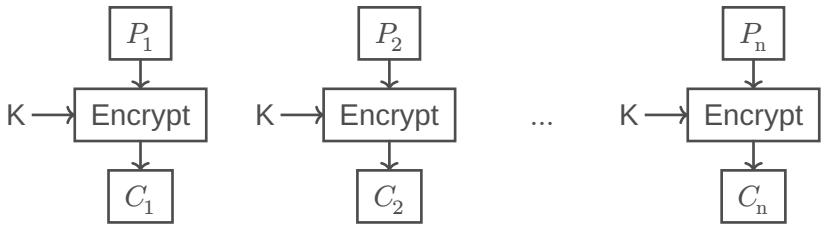
The Advanced Encryption Standard (AES) is an SP-Network

– Almost everything uses AES
– There are others (e.g. Feistel Ciphers)

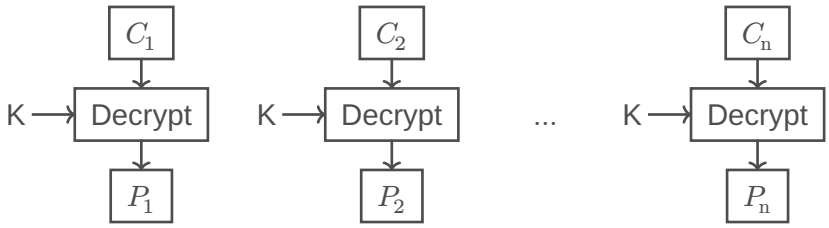
3.3.1.1. Electronic Code Block (ECB)

– Just encrypt each block one after another with same key
– Weak to redundant data divulging patterns
– Electronic codebook is not recommended!

Encryption



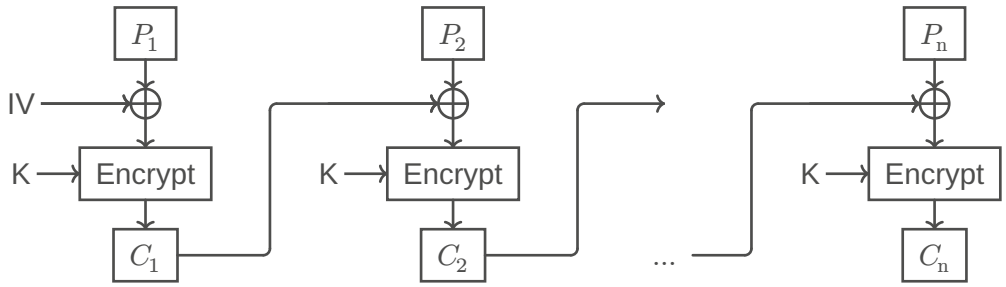
Decryption



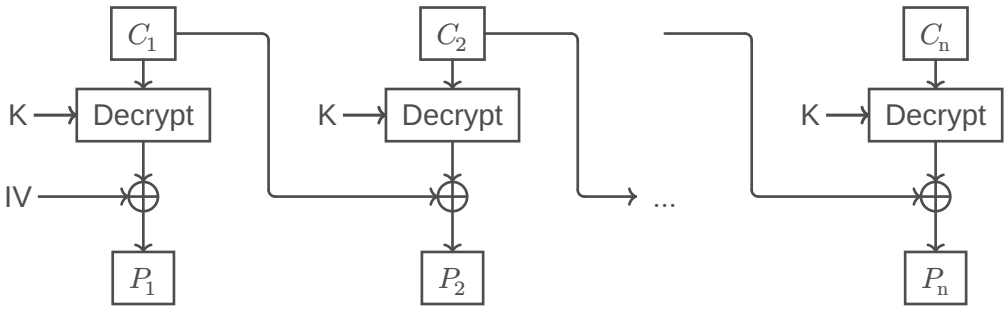
3.3.1.2. Cipher Block Chaining (CBC)

– XOR the IV with the first input, then XOR the output of each cipher block with the next input
  – Not parallelizable
  – It is better than ECB but not perfect

Encryption



Decryption



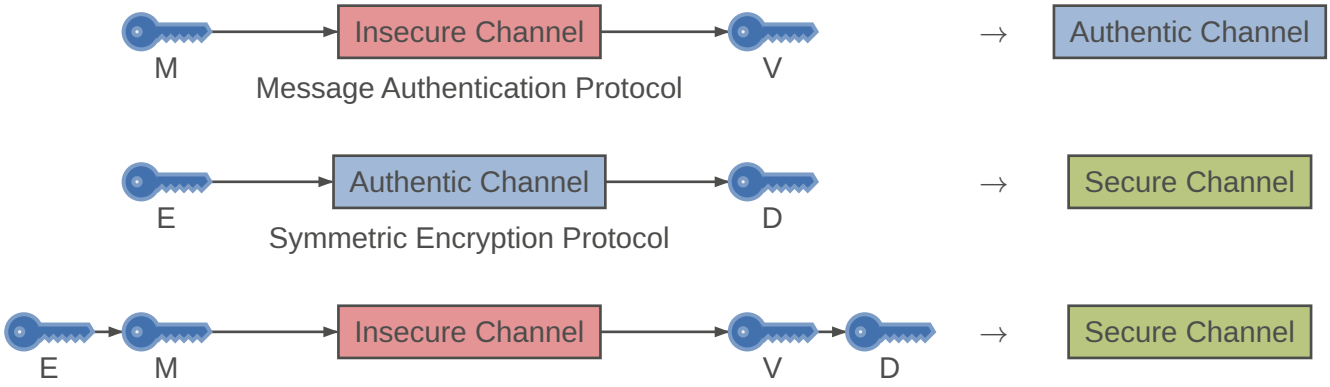
3.3.2. Stream Cipher

Inputs: Secret key + Public nonce (aka IV)

We can approximate a one-time pad by generating an infinite pseudo-random keystream

– Stream ciphers work on messages of any length (no blockification required)
– The nonce guarantees that each keystream is unique, even if the same key is reused

3.3.3. Overall Construction



Authenticity: E.g. HMAC-SHA256

Confidentiality: E.g. AES-CBC

Authenticity & Confidentiality: E.g. HMAC-SHA256-AES-CBC, AES-GCM

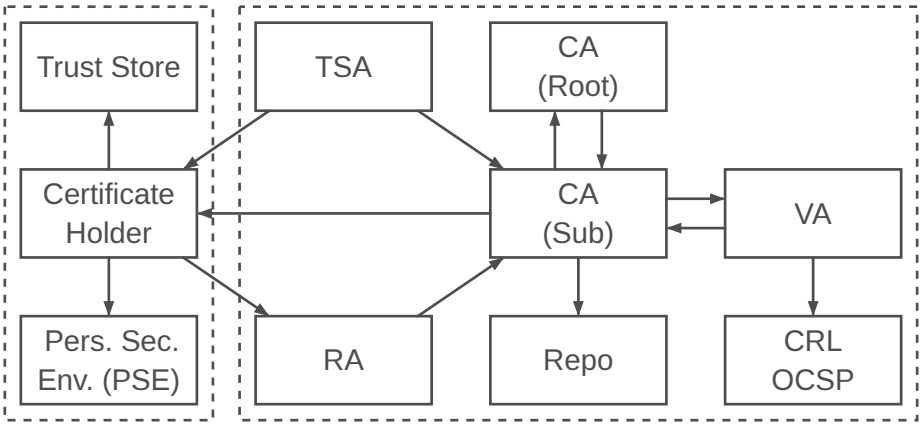
3.4. ASYMMETRIC CRYPTOGRAPHY

3.4.1. Signature Schemes

Private key used to **sign** messages. **Public key** used to **verify** signature. Signatures provide authenticity and non-repudiation. Examples: RSA, DSA, ECDSA

Signing is (often) computationally heavy. Some signature schemes natively only support small messages (naive RSA). Solution: Sign the hash of the message instead of the message itself

3.4.2. Public Key Infrastructure (PKI)



Each party has a private key and every other party knows their public key, e.g. via a registry. X.509 certificates and Certificate Authorities (CA) are used.

PKI provides an authentic channel over an insecure channel.

3.4.3. Signatures vs MACs

Signatures	***MACs***
Authenticity, non-repudiation	Authenticity only (key shared)
Slow (asymmetric)	Fast (symmetric)
Often: Use with long term keys	Often: Used with ephemeral keys

3.4.4. Establishing a shared key

Has to be established over an authentic channel.

Good pratice: Ephemeral keys, randomly generated for each session. Provides Independence between sessions and limits impact if compromised.

Problem: Authenticity, you need to know with whom we agree on a key, otherwise MitM possible. One cannot use MACs (no shared key available).

Key Exchange Protocol (KEX): Both parties contribute to the key, often the same protocol for both parties (e.g. DH KEX)

Key Encapsulation Mechanism (KEM): One party generates the key, Key is wrapped (= encrypted) and sent to the other party

> ***TODO:***
> W2 diagram slides 46

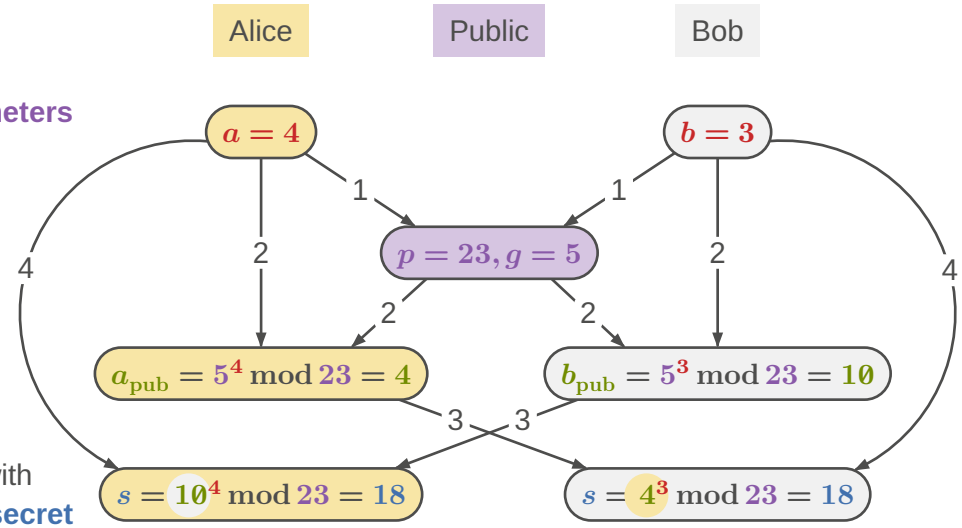
3.4.4.1. Diffie-Hellmann

– With Diffie-Hellman, two parties can jointly agree a shared secret over an insecure channel
– Every communication handshake on the internet is powered by DH
– We are exchanging some parts of the mathematical key and then we secretly create the key ourselves

Process:
– p is usually 4096 or 6144 bits
– g is a primitive root of p
– **private keys** are values between 1 and p
– The **shared secret** (often called the pre-master secret) serves as the foundation for deriving all subsequent session keys.

  - The raw shared secret is not used directly for encryption, as it is typically a very large integer (e.g., 4096 bits in RSA or DH) and may not have uniform entropy.
  - We derive a master secret using a hashed-key derivation function (HKDF), for example the SHA-256 hash function
- The only way to find **a** or **b** is to solve the Discrete Logarithm Problem.

1) Agree on **public parameters** (prime and generator)
2) Combine **private key** with **the parameters**
3) Send resulting **public keys** to each other
4) Combine other's **pubkey** with **private key** to get shared **secret**



Elliptic-Curve Diffie Hellman (ECDH) is becoming the standard nowadays due to shorter keys.

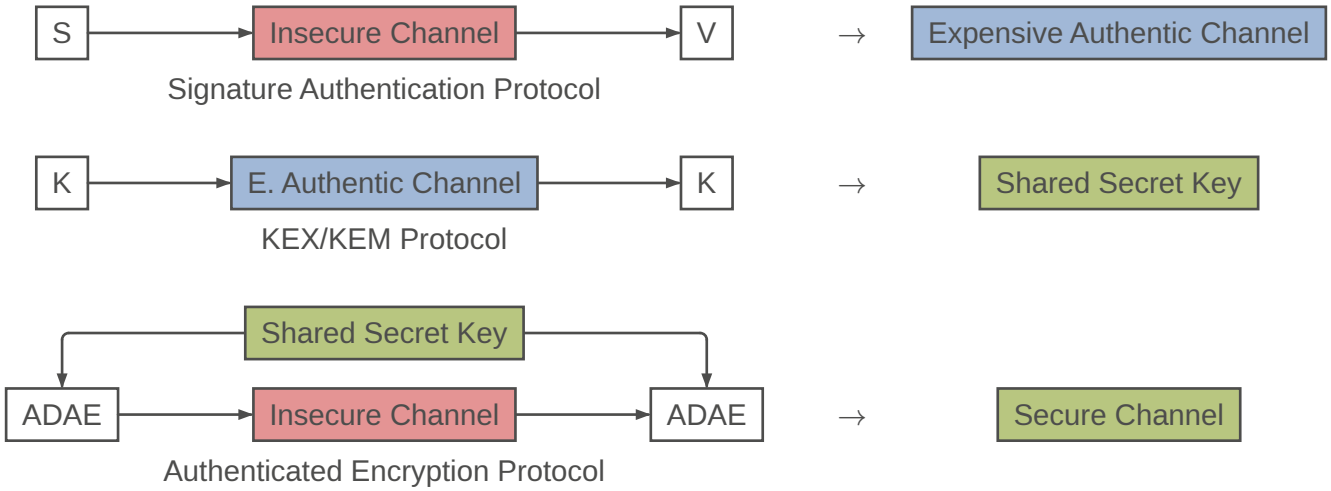
3.4.5. Asymmetric Encryption

- Public key used to encrypt messages
- Private key used to decrypt messages

E.g. RSA (factorization) or ElGamal (discrete logarithm)

TODO:
W2 slides 48,51

3.4.6. Overall Construction



3.5. MATHS <3

TODO:
W2, merge with DigCod

3.6. QUANTUM CRYPTOGRAPHY

Quantum Cryptography: Using quantum effects to do cryptography, All parties have access to quantum systems. Example: Quantum Key Exchange BB84

Post-Quantum Cryptography: Cryptography secure even if quantum computers exist, the adversary has access to a quantum computer, honest parties are (often) classic. Example: ML-KEM

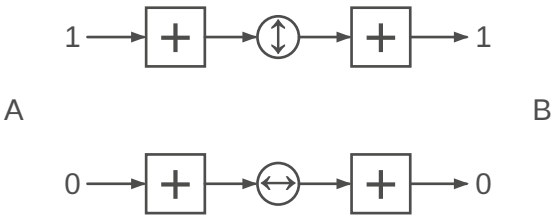
A big enough quantum computer could
– break known **asymmetric** schemes (Shor algorithm) e.g. RSA (factoring) and ElGamal, DH, ECDSA (discret log.)
– not break **symmetric** schemes e.g. Hash functions (SHA-2) and AES-256

3.6.1. Quantum Key Distribution

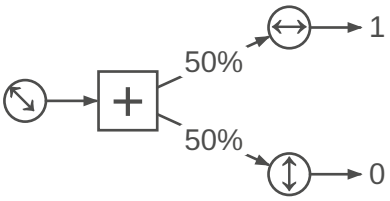
QKD's security is based on physical properties of quantum systems instead of hard math problems.

3.6.2. BB84 (By example of polarized light)

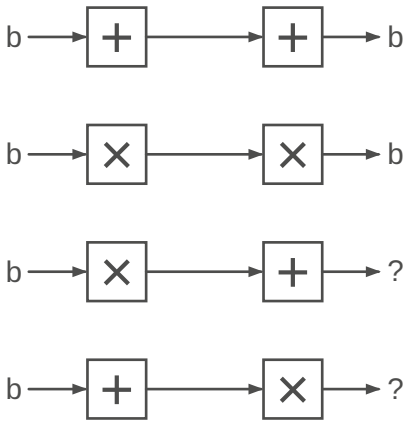
Signals sent



Randomization of output when using wrong polarization



Possible outcomes



No cloning theorem: One cannot duplicate photons. This prevents the idea of duplication and then measuring one while sending the other along.

Simplified protocol: For $1..k$:

1) Alice selects random bit b
2) Alice sends b using random base (+ or x)
3) Bob measures photon with random base (+ or x)

Bob and Alice exchange information on used bases over authentic channel
- They know which photons Bob measured correctly
- They have a common bit-string
- Errors imply listeners

3.6.2.1. Eavesdropping

Without Eavesdropping (Theoretical error rate: 0%):

50% chance Alice and Bob disagree on base	Bob gets correct bit

With Eavesdropping (Theoretical error rate: 25%):

50% chance Alice and Bob disagree on base	50% chance attacker guesses right base	
	Wrong base but Bob gets correct bit	Wrong base and Bob gets wrong bit

4. IPSEC

A collection of protocols that allow secure communication over IP networks. Provides confidentiality and/or authenticity and integrity of (parts of) IP packets.

4.1. PROTOCOLS

A collection of protocols that allow secure communication over IP networks:

Security Architecture for the Internet Protocol: RFC 4301 Defines basic architecture and requirements for IPsec gateways. In particular: Security Associations Management, e.g. “MUST support ESP”, z.B. “MAY support AH”

Authentication Header (AH): RFC 4302 Protocol to protect the integrity/authenticity of both payload and header data

Encapsulating Security Payload (ESP): RFC 4303 Protocol to protect the confidentiality and integrity/authenticity of payload data

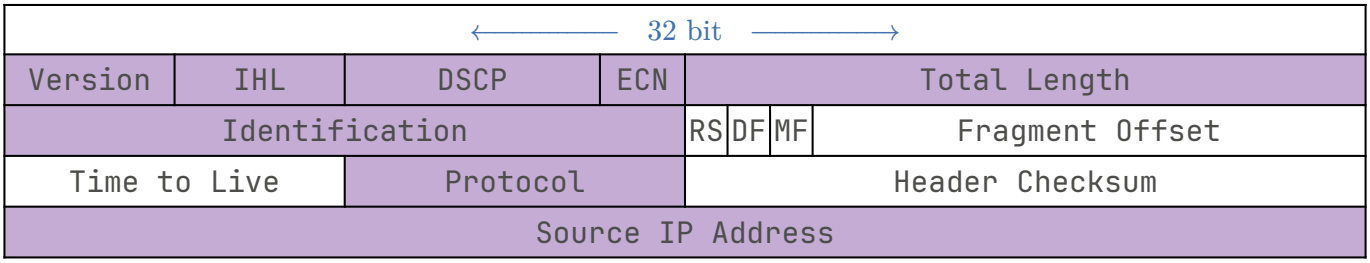
Internet Key Exchange v2 (IKEv2): RFC 7296 Protocol for manual and automated key exchange
Relevant for creating Security Associations used in AH and ESP

4.1.1. Authentication Header (AH)

Protects the integrity/authenticity of the payload and parts of the IP header (purple). Provides (optional) replay protection, does not provide confidentiality.

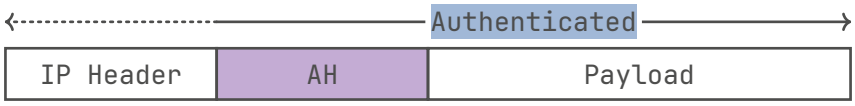
Protocol Field in the IP header: 0x33

Protected IP Header:



← 32 bit →
Destination IP Address
Options (if IHL > 5)

IP Packet with AH:



Authentication Header:

← 32 bit →		
Next Header	Payload Len	RESERVED
Security Parameters Index (SPI)		
Sequence Number Field		
Integrity Check Value - ICV (variable)		

Field	***Definition***
Next Header (8 bit)	ID of the protocol of the payload (eg. TCP/UDP)
RESERVED (16 bit)	All zeroes
Security Parameters Index (SPI) (32 bit)	Identifies the Security Association (SA), which in turn defines security attributes like Keys and Authentication algorithms
Sequence Number Field (32 bit)	For replay protection
Integrity Check Value - ICV (variable) (32 bit)	e.g. HMAC_SHA2_512_256

The sender increments the sequence number by one for every packet sent., the receiver stores the highest sequence number received so far.
– Too old or already received packets are rejected
– “Too old” is defined via the anti-replay window (default: 64)

The sequence number (32-bit value) is tracked per security association (SA)

– ***TODO:***
max 232-1 packets can be sent per SA

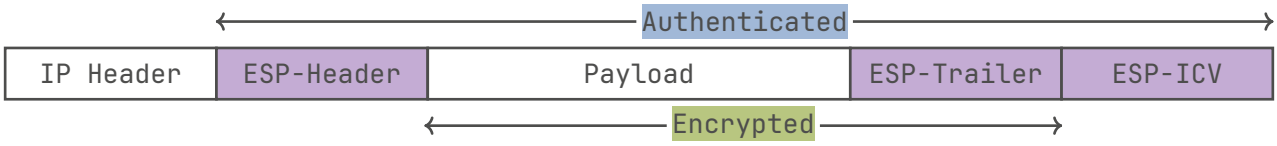
– Periodically a new SA is built

Extended Sequence Numbering are 64-bit values used for high-speed connections (>100 Gbps)

4.1.2. Encapsulating Security Payload (ESP)

Provides Confidentiality and authenticity of the payload. Authenticity (via ICV) and replay protection is optional (but recommended).

The IP header is not protected. Protocol Field in the IP header: 0x34



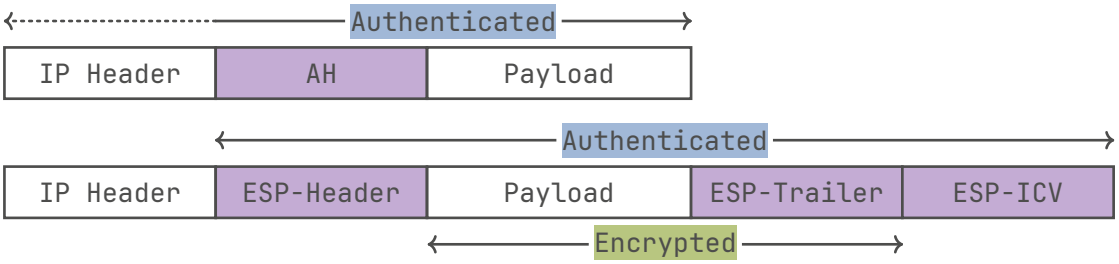
TODO:
W4 S14

4.2. MODES OF OPERATION

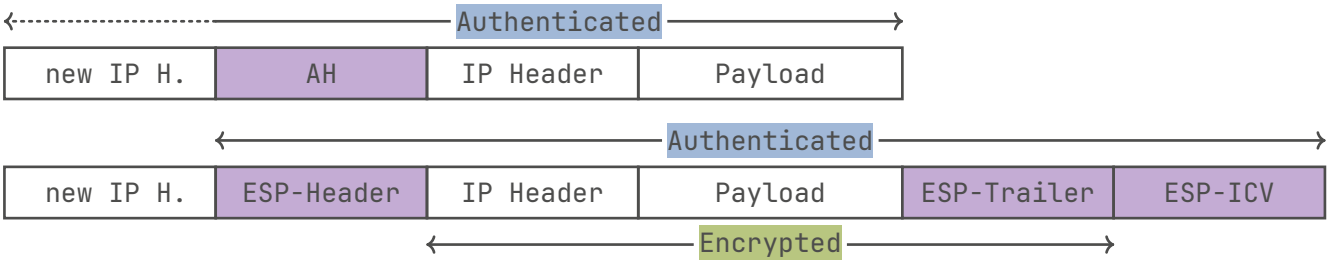
4.2.1. Transport vs. Tunnel Mode

Both AH and ESP have two modes of operation:

Transport Mode: Protects IP packets directly



Tunnel Mode: IP packets are packed into a new IP packet (as payload) which is then protected



4.2.2. Transport Mode

– The IP traffic is intended for the IPsec endpoints
– End-to-end security, but transparent for applications
– ESP: secure channel
– AH: authentic channel

4.2.3. ESP Tunnel Mode

Site-to-Site VPN:

– Tunnel between two IPsec gateways
– Transparent for endpoints. Independent of application
– Internal IP addresses remain confidential
– E.g. VPN tunnel between corporate networks

Client-to-Site VPN:

– Tunnel between an IPsec client and an IPsec gateway
– Internal IP addresses remain confidential
– E.g. Remote access to corporate network

4.3. SECURITY ASSOCIATIONS

For IPsec communication the peers need to share information on
– Cipher Suite, Key Length, Keys, …
– Protocol: AH or ESP
– Mode: Transport or Tunnel

– Sequence number counter
– Validity period of security associations

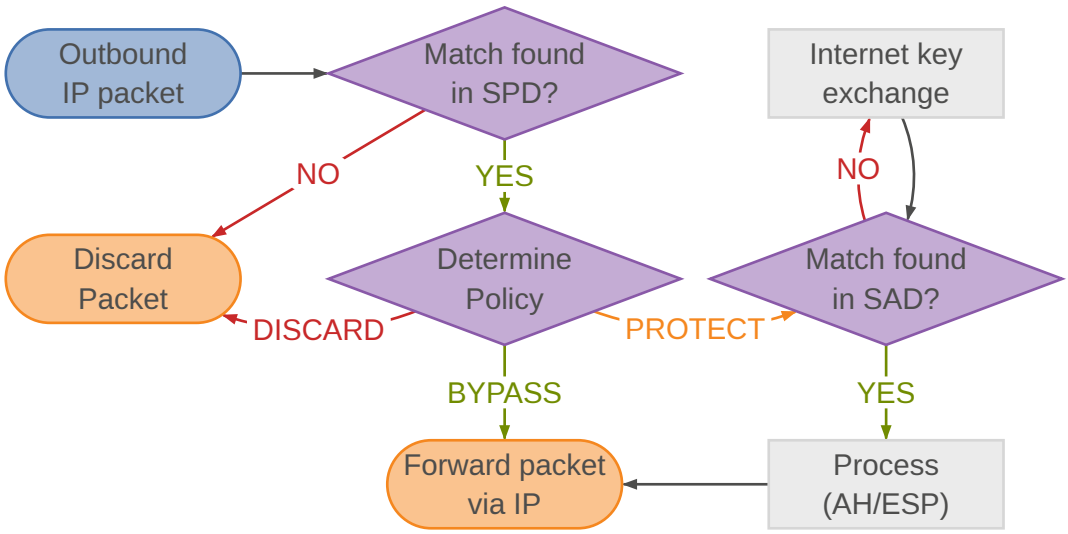
This information is stored in a security association (SA), Identified by a Security Parameters Index (SPI). Separate SAs for each communication direction, i.e. one per sender, and separate SAs for AH and ESP.

4.3.1. SAD and SPD

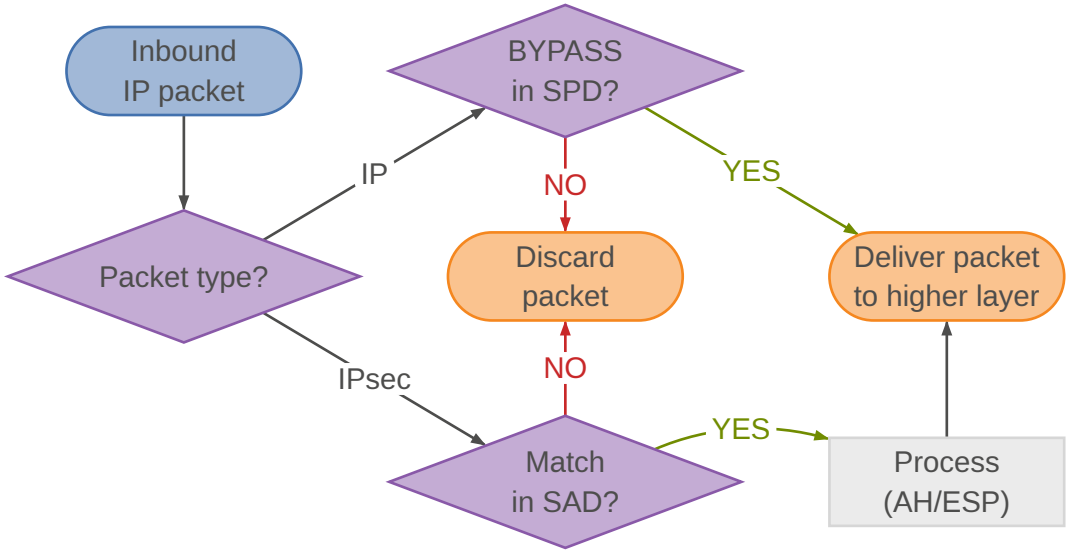
Security Policy Database (SPD): decides how to handle an IP packet
- Decision: Discard, Bypass, Protect
- Based on selectors: IP addresses, Next Layer Protocol, …

Security Association Database (SAD): holds all the security associations
- Incoming packets: SPI and possibly IP address point to SAD entry
- Outgoing packets: SPD entry points to SA

4.3.2. Handling of outbound packets



4.3.3. Handling of inbound packets



4.4. INTERNET KEY EXCHANGE V2 (IKEV2)

Protocol for mutual authentication and building of security associations between an Initiator and a Responder. RFC 7296

Goal: Initiator and Responder each have
– a security association for IKE transport (IKE-SA)
– security associations for ESP or AH traffic (CHILD-SA)

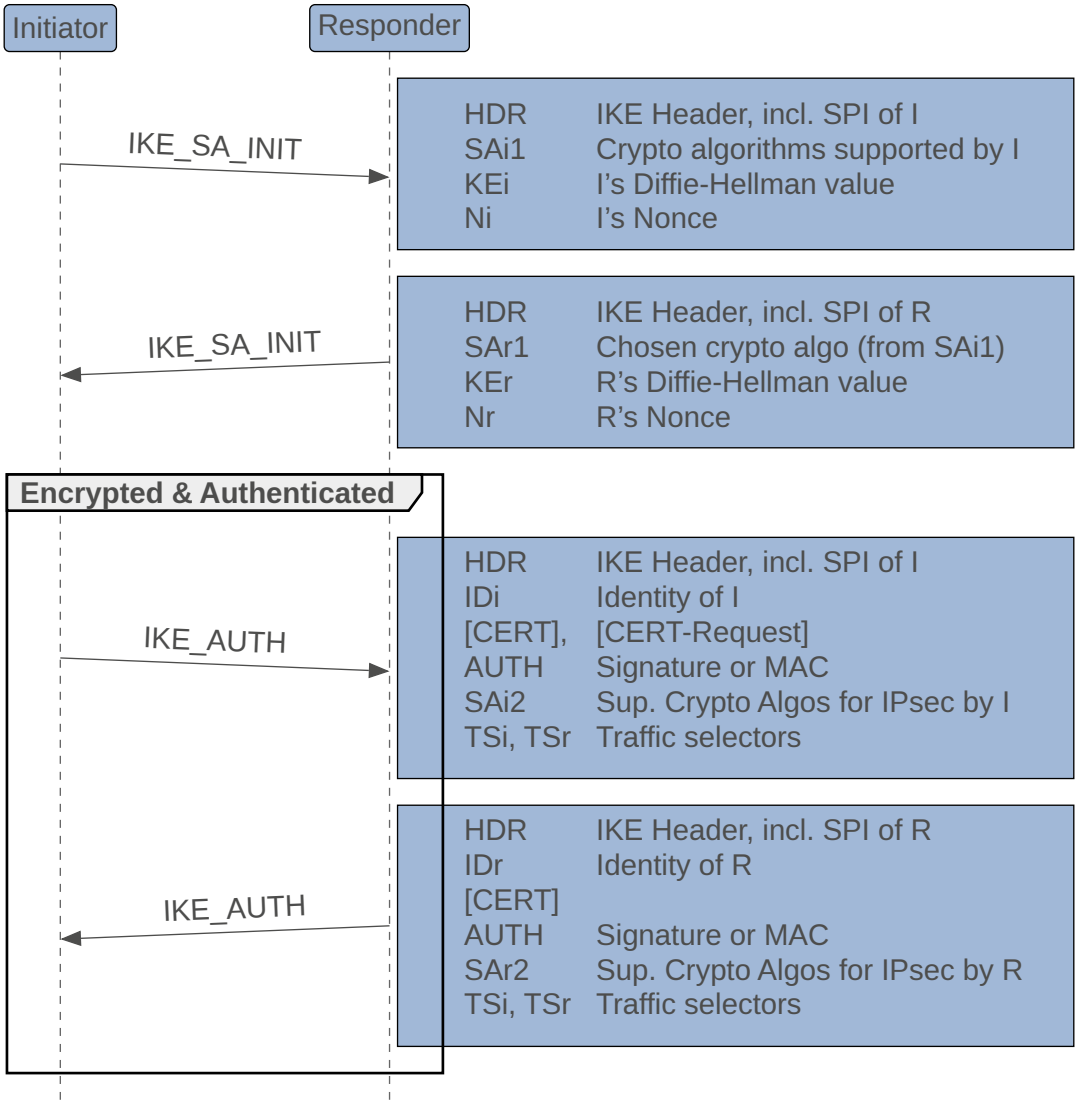
IKE messages are transmitted over UDP (port 500).

Protocol Phases:



4.4.1. Initial Exchanges

– The first two messages (IKE_SA_INIT) negotiate the security parameters and do a DH key exchange.
– The second pair of messages (IKE_AUTH) verifies the identities and establishes a first CHILD_SA.



4.4.2. AUTH

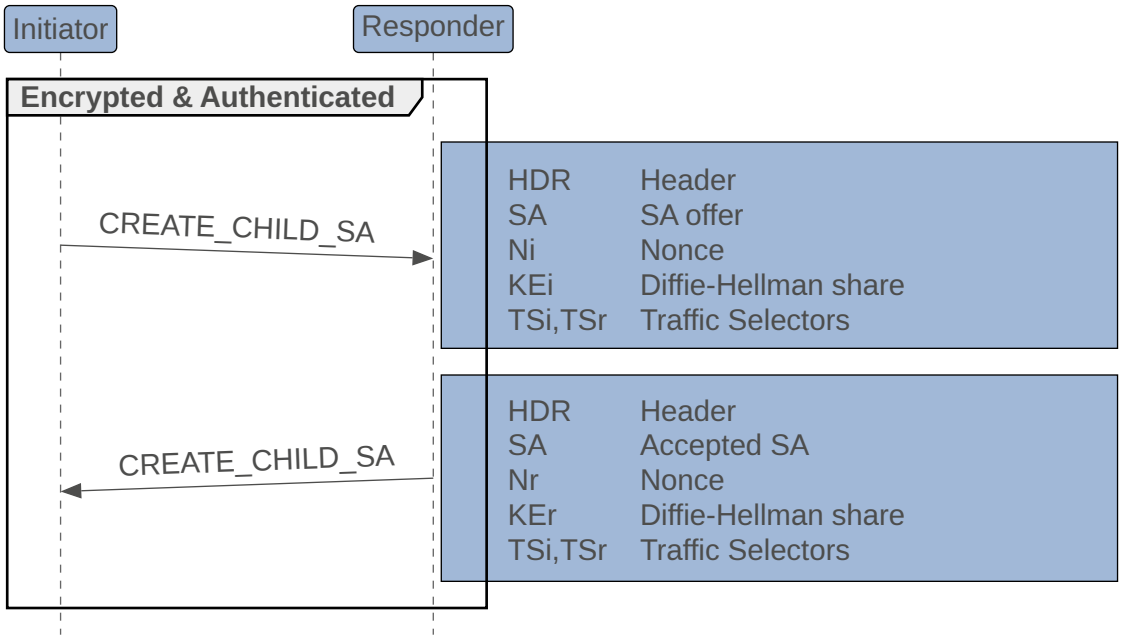
In the IKE_AUTH messages

– The identity of peers is verified by means of
  – Signatures (Certificates are sent in the messages)
  – Symmetric MAC if using a pre-shared-key
– The authenticity of the IKE_INIT messages is verified as well

– Instead of AUTH one can use Extensible Authentication (EAP)
  – Initiator omits the AUTH field to indicate this mode
  – Normally used to authenticate an initiator
  – Uses an additional exchange IKE_AUTH messages

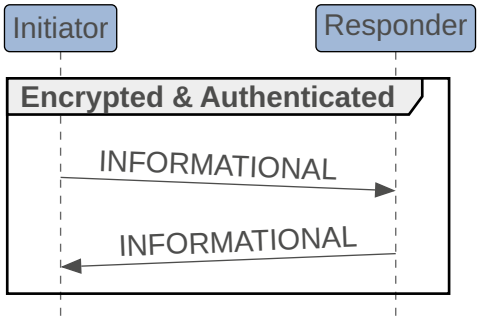
4.4.3. Child SA Exchange

Happens on new Child-SA or Rekeying of a Child-SA or Rekeying of the IKE-SA. Based on existing key material, optionally rekeying using DH.



4.4.4. INFORMATIONAL

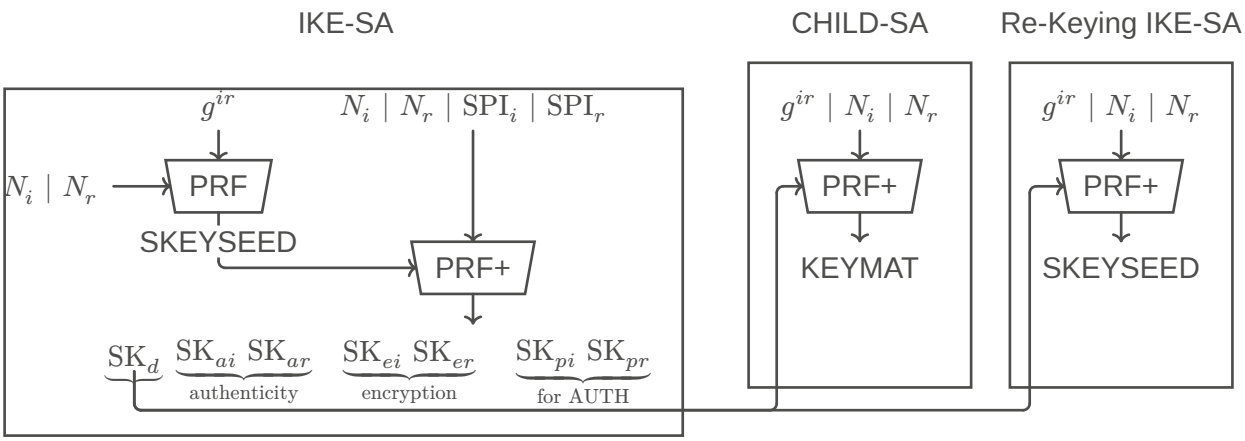
For Control messages, e.g. error messages, event notification, deletion of SAs.



4.4.5. Key Derivation (Function)

PRF: Pseudo Random Function

IKEv2 uses a PRF to derive key material, the PRF depends on cipher suite, e.g. AES-CMAC. Inputs to the PRF comes from Existing key material, the Nonce, and DH key exchange (ephemeral key).



TODO:
W4 S36

4.4.6. Algorithms

IPsec use various cryptographic algorithms (in IKEv2 negotiated (SAs)). RFC 8247 specifies “mandatory-to-implement” algorithms for IKEv2. RFC 8221 specifies algorithms for AH. Various RFCS define cipher suites with the goal of interoperability and a certain level of security, like RFC 4308 : Cryptographic Suites for IPsec (2005).

4.5. IPSEC & NAT

In one-to-one NAT, the NAT:
– replaces source address with its public address
– only works with one client

In Port Address Translation (PAT), the NAT:
– replaces source address with its public address
– Replaces (UDP/TCP) source port with unique free port
– Relays incoming packets to the right client based on the port number.

4.5.1. Problems

AH is incompatible as IP addresses are authenticated.

In ESP:
– PAT: The payload (UDP/TCP) is encrypted and cannot be changed by the NAT
– One-to-One NAT: TCP (and UDP in IPv6) has checksums that break when swapping out IP

In IKE:
– Using IP addresses as an identifier (IKEv1) causes issues.
– IKE responder needs to accept packets from a non 500 port.

4.5.2. NAT-T and IPsec

RFC 3948 : UDP-Encapsulation of IPsec ESP Packets
– UDP-Header between IP-Header and ESP-Header
– Source und destination port: 4500



RFC 4306 : IKEv2 can (implementation optional)
– Detect NAT (NAT_DETECTION_SOURCE_IP and NAT_DETECTION_DESTINATION_IP in INIT-msg)

– Negotiate UDP-Encapsulation for IKE and ESP
– Source und dst. port: 4500, reply to arbitrary port



5. WIREGUARD

Wireguard is a Fast, Modern, Secure, Free, Open Source, Best, Bestest VPN Tunnel.

5.1. CRYPTO STACK

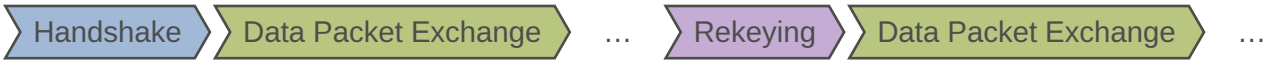
Symmetric:
– ChaCha20 for symmetric encryption, authenticated with Poly1305, using RFC 7539 ‘s AEAD construction
– BLAKE2s for hashing and keyed hashing, described in RFC 7693
– SipHash24 for hashtable keys
– HKDF for key derivation, as described in RFC 5869

Asymmetric:
– Curve25519 for ECDH, the public and private keys are for DH

5.2. SETUP

– Both parties generate a DH key pair.
– Exchange public keys with each other (done by an admin not the WireGuard protocol).
– Each party has a config file that maps the other party’s public key to their AllowedIPs
– When connecting a handshake of two messages is exchanged to verify the identity of the other party

5.3. PROTOCOL PHASES



5.3.1. Connection establishment

WireGuard uses the Noise_IK handshake pattern of the Noise Protocol Framework

I (Immediate): The initiator’s own static public key is transmitted (encrypted) inside Message 1, so the responder learns the initiator’s identity during the handshake itself.

K (Known): The initiator already knows the responder’s static public key before the handshake begins, because both sides configured each other’s public keys out-of-band as part of setup.

5.3.2. First Handshake Message

Initiator sends a message to the responder with:

Ephemeral (public key): Initiator’s one-time key for this session, in plaintext.

Encrypted Static (public key): Initiator’s long-term static public key, encrypted under a combination of the initiator’s ephemeral key and the responder’s pre-known static public key.

Encrypted Timestamp (TAI64N): 12 Byte, for replay protection

5.3.3. Second Handshake Message

Responder sends a message to the responder, after processing the first message, with:

Ephemeral (public key): Responder's one-time key for this session, in plaintext.

Encrypted Empty (payload): Proof of key possession → Encrypting an empty payload forces the responder to demonstrate it has correctly derived the same session key material as the initiator, without transmitting any additional identifying information.

5.3.4. Key Derivation

Conceptually a combination of DH and KDF as in IKE, but the overall design simpler than IKE.
– Long-term keys are used for authentication
– Ephemeral keys provide forward secrecy

Encryption: One key for each direction of communication

Post-quantum: Add pre-shared key into the mix

5.4. IKEV2/IPSEC VS WIREGUARD

Aspect	*IKEv2/IPSEC*	*WireGuard*
Connectivity	Connection-oriented	Connectionless
On Top Of	IP (alt. UPD)	UDP
Configuration Complexity	Medium	Low (static keys)
Configuration Possibilities	Wide	Limited
Speed	Fast	Generally faster
Quantum Resistance	Yes (Right Config)	Yes (with PSK enabled)
Which one to pick?	Frequent Mobile Roaming, Enterprise and Legacy Hardware, Regulatory Compliance	High-Performance Streaming and Gaming, Simple DIY and Self-Hosted VPNs, Resource-Constrained Hardware